

JIACHANG ZHANG

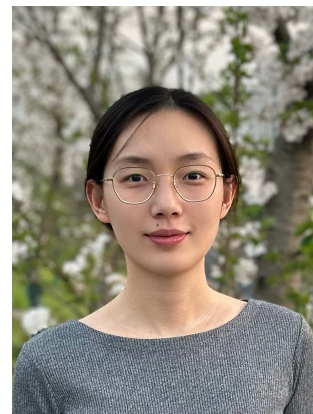
School of Astronomy and Space Science, Nanjing University, China

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Date of Birth: 11st, March, 2001

Gender: Female



EDUCATION

Nanjing University, Nanjing, China

School of Astronomy & Space Science, Ph.D. candidate, Astrophysics

Sep 2022 - June 2027 (expected)

Supervisor: Prof. Zhiyuan Li

Nanjing University, Nanjing, China

Kuangyaming Honors School, B.S., Astronomy

Sep 2018 - June 2022

RESEARCH INTERESTS

Time-domain high-energy astrophysics; X-ray binaries and accreting compact objects; X-ray timing and variability; Compact-object populations in nearby galaxies; Star clusters and their X-ray populations; Diffuse ionized gas and stellar feedback

ACADEMIC VISIT

University of Washington, Seattle, WA, US

Sep 2024 - Dec 2024

Supervisor: Dr. Benjamin F. Williams

Ionized gas and related star populations in M31 and M33

SELECTED PRESENTATIONS

The first Shen Kuo Symposium of Chinese Astronomical Society, Wuhan, Hubei, China, Jan 2024.

“A *Chandra* Search for Periodic X-ray Sources in the Bulge of M31 ” (*Poster Presentation*)

25 Years of Chandra Science Symposium, Boston, Massachusetts, USA, Dec 2024.

“A *Chandra* Search for Periodic X-ray Sources in the Bulge of M31” (*Poster Presentation*)

The Second Shen Kuo Symposium of the Chinese Astronomical Society, Xiamen, Fujian, China, Dec 2024.

“Searching for Periodic X-ray Sources in the Bulges of M31, M81, and Centaurus A” (*Oral Presentation*)

The 2025 Workshop on the Structure, Formation, and Evolution of Stellar Clusters, Dali, Yunnan, China, Mar–Apr 2025.

“X-ray Emission from Star Clusters in M31 and M33” (*Oral Presentation*)

The 2025 Annual Science Meeting and Early Science Workshop of the Chinese Space Station Survey Telescope (CSST), Kunming, Yunnan, China, Jul 2025.

“A High-Resolution Study of H α -emitting Sources in M33” (*Oral Presentation*)

The Third Shen Kuo Symposium of the Chinese Astronomical Society, Zhuhai, Guangdong, China, Jan 2026.

“Probing Periodic and Aperiodic Variability of X-ray Sources in M31, M81, and Centaurus A with *Chandra*” (*Oral Presentation*)

Frontiers in Diffuse Hot Gas and Space-based X-ray Telescopes, Suzhou, Jiangsu, China, Jun 2026.

“Recurrent Luminous X-ray Flares from a Persistent X-ray Binary in M81” (*Oral Presentation*)

46th COSPAR Scientific Assembly, Florence, Italy, Aug 2026.

“Beyond the Milky Way: Systematic X-ray Timing of Extragalactic LMXBs in M31, M81 and Cen A with the Chandra Archive” (*Oral Presentation*)

PUBLICATIONS

Zhang, J., et al. 2026, “An Unusual Persistent X-ray Source in M81 Exhibiting Recurrent Luminous Flares,” to be submitted.

Zhang, J., Li, Z., Hua, Z., & Bao, T. 2026, “Probing periodic and aperiodic variability of X-ray sources in M31, M81, and Centaurus A with Chandra,” *MNRAS*, **546**, staf2237. doi: 10.1093/mnras/staf2237.

Ma, Q.-Q.^{^†}, **Zhang, J.**^{^†}, Gu, W.-M., Li, Z., Weng, S.-S., & Bao, T. 2026, “The first extragalactic ultra-compact X-ray binary: A candidate black hole-white dwarf system,” *The Innovation*, **7**, 101181. doi: 10.1016/j.xinn.2025.101181.

Li, Z., Su, Z., Wang, S., Zhou, Y. F., Li, Z., Lyu, X., Meng, F., Zhang, K., Li, D., Tsai, C., Wu, J., & **Zhang, J.** 2025, “Ring or No Ring — Revisiting the Multiphase Nuclear Environment in M31,” *ApJ*, **983**, 182. doi: 10.3847/1538-4357/adc103.

Zhang, J., Bao, T., & Li, Z. 2024, “A Chandra search for periodic X-ray sources in the bulge of M31,” *MNRAS*, **530**, 2096–2113. doi: 10.1093/mnras/stae1002.

PROPOSALS

Co-Investigator, “Joint MCI–IFS Observations of M31,” CSST Early Science Program.

Co-Investigator, “Joint IFS–MCI Observations of the Virgo Cluster,” CSST Early Science Program.

Co-Investigator, “Joint MCI–IFS Observations of M33,” CSST Early Science Program.

MAIN AWARDS AND HONORS

2024 — Tiangong International Exchange Scholarship, Nanjing University ($\approx$ CNY 80,000)

2024 — Outstanding Poster Award, The First Shen Kuo Symposium of the Chinese Astronomical Society, Wuhan, Hubei, China

“A Chandra Search for Periodic X-ray Sources in the Bulge of M31”

2022 — Lu Dexin Award for Outstanding Graduates, Kuang Yaming Honors School, Nanjing University

2020 — Meritorious Winner, Mathematical Contest in Modeling (MCM), COMAP

2020 — People's Scholarship, Nanjing University

2020 — Scholarship for Top-Talent Students, Nanjing University

2019 — People's Scholarship, Nanjing University

2019 — Scholarship for Top-Talent Students, Nanjing University

TEACHING EXPERIENCE

Course: *An Introduction to X-ray Astronomy*, Spring Semester, Department of Astronomy, Nanjing University

Teaching Assistant and Lecturer *Feb 2026 – July 2026*

RESEARCH MENTORING

2025-2026 — Undergraduate Research Mentor, Undergraduate Innovation and Entrepreneurship Training Program, Nanjing University

“An X-ray Study of Massive Stars in the Solar Neighborhood” — Mentored a project using archival Chandra and Gaia data to identify and characterize long-duration stellar X-ray flares and investigate their statistical properties.

2026-present — Undergraduate Research Mentor, Nanjing University

“Spectral Evolution and State Variability of X-ray Binaries in M31” — Supervising a project using archival X-ray observations to investigate spectral evolution, luminosity-dependent behavior, and possible accretion-state transitions in M31 X-ray binaries.

SKILLS

X-ray Data Analysis: Extensive experience in the reduction and analysis of data from **Chandra**, **XMM-Newton**, **Swift**, and **NuSTAR**, including source detection, spectral extraction and modelling, light-curve analysis, periodic and aperiodic timing analysis, and searches for transient and variable sources.

Optical Imaging Analysis: Experience with **Hubble Space Telescope (HST)** imaging for point-source photometry and the analysis of diffuse ionized gas and emission-line sources; participant in the early-science program of the **China Space Station Telescope (CSST)**.

Astronomical Software: CIAO, SAS, HEASoft, XSPEC, DS9, DOLPHOT, Astropy and so on.

Programming and Scientific Computing: Python, IDL, Bash/Shell, and C++; LaTeX for scientific writing and typesetting.

Languages: Mandarin Chinese (native); English (professional working proficiency).

OUTREACH AND ACADEMIC SERVICE

2023 — **Volunteer**, Undergraduate Summer Camp, School of Astronomy and Space Science, Nanjing University

2024 — Student Speaker, Undergraduate Summer Camp, School of Astronomy and Space Science, Nanjing University

Gave an invited talk sharing experiences of graduate study, research, and student life with prospective students.

2023–2025 — Volunteer, Astronomy Night Public Outreach Program, Nanjing University

PRESS RELEASE

Science and Technology Daily - “Scientists Discover First Extragalactic Ultra-Compact X-ray Binary” - April 2026

https://www.stdaily.com/web/gdxw/2026-04/06/content_498156.html

<https://astro.xmu.edu.cn/info/1050/18705.htm>